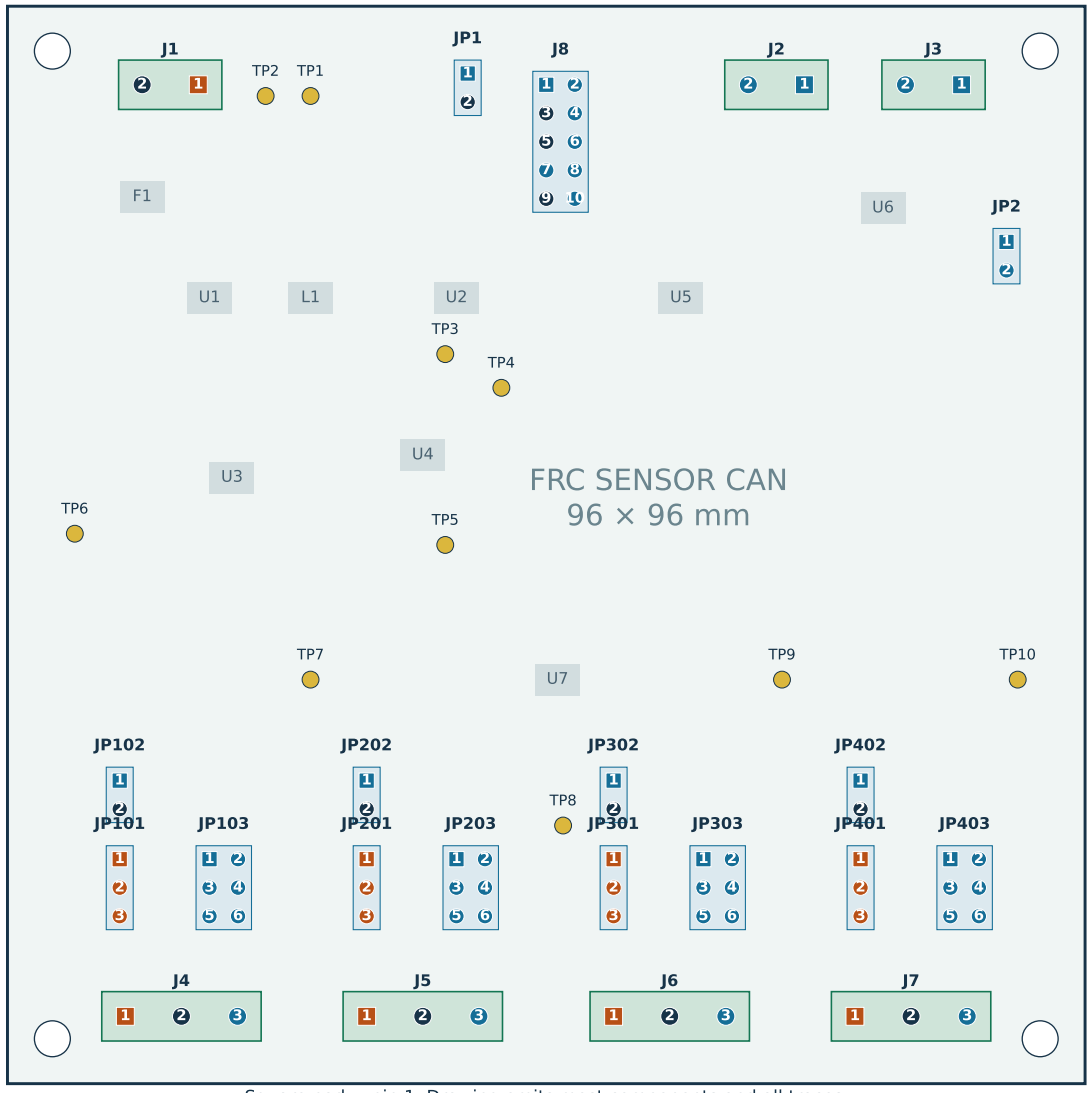
Bench connector map

TOP view · input/CAN edge at top · sensor edge at bottom · pad positions taken from the PCB



Square pad = pin 1. Drawing omits most components and all traces.

J1 · board input

Pin 1 = +6–18 V target; pin 2 = GND.
In this top view, + is RIGHT and GND is LEFT.
Use one protected robot distribution branch.

J2 / J3 · CAN trunk

Pin 1 = CAN H; pin 2 = CAN L.
In this top view, H is RIGHT and L is LEFT.
JP2 adds 120 Ω. Leave open at an intermediate node.
JP1 shorts RESET to GND; normally leave open.

J4 / J5 / J6 / J7 · channels 1 / 2 / 3 / 4

LEFT → RIGHT: 1 POWER · 2 GND · 3 SIGNAL.
POWER is selected 5 V or protected battery.
300 mA target per shared rail, not per connector.
External sensor supply: leave POWER disconnected.

J8 · custom service header, 2.54 mm pitch

Top row:	1 VTref OUT	2 SWDIO
	3 GND	4 SWCLK
	5 GND	6 UART TX
	7 BOOT0	8 UART RX
Bottom row:	9 GND	10 RESET

Use mapped individual leads; not a Cortex ribbon.
3.3 V UART only. Leave adapter power disconnected.

Test points

TP1 GND · TP2 protected input · TP3 internal 5 V
TP4 3V3 · TP5 switched 5 V · TP6 switched battery
TP7 / TP8 / TP9 / TP10 = conditioned inputs 1–4

Jumper examples · channel 1

Disconnect board and sensor power before changing shunts. These drawings keep the TOP-view pin orientation.

Industrial NPN



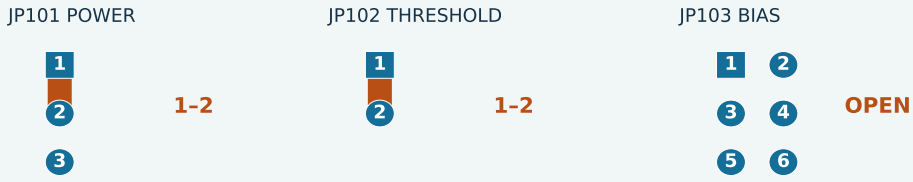
Battery-capable sensor; selected-port pull-up.

Industrial PNP



Battery-capable sensor; signal pull-down.

5 V push-pull



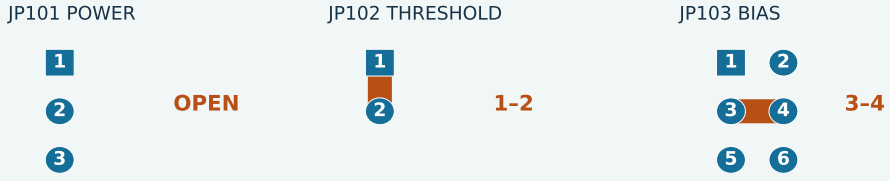
5 V supply; externally driven 3.3/5 V signal.

3.3 V open-drain signal



Only if the sensor supply accepts 5 V.

Dry contact to ground



Contact between SIGNAL and GND; POWER unused.

Externally powered 24 V PNP



Common GND and SIGNAL only; POWER unused.

Repeat JP1xx settings at JP2xx / JP3xx / JP4xx for channels 2 / 3 / 4. Fit at most ONE power shunt and ONE bias shunt per channel.

Threshold OPEN = industrial; FITTED = logic. Verify actual sensor LOW/HIGH/leakage limits and inversion. Firmware defaults to sensor rails OFF.

REV A PROTOTYPE | Unbuilt; follow the handbook and qualification procedure.